

#In this code we will demonstrate the lemma.7.7 • We will check that the limit L by $\frac{b}{a}$ where b goes to infinity is different for 1 and 0.

#Define r3

$$\begin{aligned}
 r3 := & (a^{22} b^{10} + (-6 b^{11} + 8 b^9) a^{21} + (15 b^{12} + 6 b^{10} + 30 b^8) a^{20} + (-18 b^{13} - 102 b^{11} + 162 b^9 \\
 & + 74 b^7) a^{19} + (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^{18} + (24 b^{15} - 96 b^{13} - 762 b^{11} \\
 & + 1884 b^9 + 1380 b^7 + 252 b^5) a^{17} + (-294 b^{16} - 1740 b^{14} - 3162 b^{12} - 2920 b^{10} + 1908 b^8 \\
 & + 1092 b^6 + 145 b^4) a^{16} + (24 b^{17} - 132 b^{15} - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 \\
 & + 1380 b^5 + 74 b^3) a^{15} + 3 b^2 (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 \\
 & + 636 b^4 + 200 b^2 + 10) a^{14} + (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} - 34416 b^{11} \\
 & - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^{13} + (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} \\
 & - 67416 b^{12} - 68232 b^{10} - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{12} - 6 b (b^{20} + 17 b^{18} \\
 & + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 + 17 b^2 + 1) a^{11} \\
 & + b^2 (b^{20} + 6 b^{18} + 93 b^{16} - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 \\
 & - 3162 b^4 + 198 b^2 + 15) a^{10} + (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} - 8268 b^{13} - 34416 b^{11} \\
 & - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^9 + (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} \\
 & - 24921 b^{12} - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^8 + (74 b^{19} + 1380 b^{17} + 4692 b^{15} \\
 & + 3340 b^{13} - 8010 b^{11} - 2472 b^9 - 132 b^7 + 24 b^5) a^7 + (145 b^{18} + 1092 b^{16} + 1908 b^{14} \\
 & - 2920 b^{12} - 3162 b^{10} - 1740 b^8 - 294 b^6) a^6 + (252 b^{17} + 1380 b^{15} + 1884 b^{13} - 762 b^{11} \\
 & - 96 b^9 + 24 b^7) a^5 + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^4 + (74 b^{15} + 162 b^{13} \\
 & - 102 b^{11} - 18 b^9) a^3 + (30 b^{14} + 6 b^{12} + 15 b^{10}) a^2 + (8 b^{13} - 6 b^{11}) a + b^{12} :
 \end{aligned}$$

#Calculate the partial derivative by r3 in with respect to a

Dar3 := diff(r3, a)

$$\begin{aligned}
 \text{Dar3} := & 22 a^{21} b^{10} + 21 (-6 b^{11} + 8 b^9) a^{20} + 20 (15 b^{12} + 6 b^{10} + 30 b^8) a^{19} + 19 (-18 b^{13} - 102 b^{11} + 162 b^9 + 74 b^7) a^{18} \\
 & + 18 (3 b^{14} + 198 b^{12} + 93 b^{10} + 600 b^8 + 145 b^6) a^{17} + 17 (24 b^{15} - 96 b^{13} - 762 b^{11} + 1884 b^9 + 1380 b^7 + 252 b^5) a^{16} \\
 & + 16 (-294 b^{16} - 1740 b^{14} - 3162 b^{12} - 2920 b^{10} + 1908 b^8 + 1092 b^6 + 145 b^4) a^{15} + 15 (24 b^{17} - 132 b^{15} - 2472 b^{13} - 8010 b^{11} + 3340 b^9 + 4692 b^7 \\
 & + 1380 b^5 + 74 b^3) a^{14} + 42 b^2 (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 + 636 b^4 + 200 b^2 + 10) a^{13} \\
 & + 13 (-18 b^{19} - 96 b^{17} - 2472 b^{15} - 14520 b^{13} - 34416 b^{11} - 8268 b^9 + 3340 b^7 + 1884 b^5 + 162 b^3 + 8 b) a^{12} \\
 & + 12 (15 b^{20} + 198 b^{18} - 3162 b^{16} - 26352 b^{14} - 67416 b^{12} - 68232 b^{10} - 24921 b^8 - 2920 b^6 + 93 b^4 + 6 b^2 + 1) a^{11} \\
 & - 66 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 + 17 b^2 + 1) a^{10} \\
 & + 10 b^2 (b^{20} + 6 b^{18} + 93 b^{16} - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 - 3162 b^4 + 198 b^2 + 15) a^9 \\
 & + 9 (8 b^{21} + 162 b^{19} + 1884 b^{17} + 3340 b^{15} - 8268 b^{13} - 34416 b^{11} - 14520 b^9 - 2472 b^7 - 96 b^5 - 18 b^3) a^8 \\
 & + 8 (30 b^{20} + 600 b^{18} + 1908 b^{16} - 3516 b^{14} - 24921 b^{12} - 26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^7 \\
 & + (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} - 8010 b^{11} - 2472 b^9 - 132 b^7 + 24 b^5) a^6 + (145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12} - 3162 b^{10} - 1740 b^8 - 294 b^6) a^5 \\
 & + (252 b^{17} + 1380 b^{15} + 1884 b^{13} - 762 b^{11} - 96 b^9 + 24 b^7) a^4 + (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^3 + (74 b^{15} + 162 b^{13} - 102 b^{11} - 18 b^9) a^2 \\
 & + (30 b^{14} + 6 b^{12} + 15 b^{10}) a + (8 b^{13} - 6 b^{11}) :
 \end{aligned} \tag{1}$$

$$\begin{aligned}
& -26352 b^{10} - 11586 b^8 - 1740 b^6 + 3 b^4) a^7 + 7 (74 b^{19} + 1380 b^{17} + 4692 b^{15} + 3340 b^{13} \\
& - 8010 b^{11} - 2472 b^9 - 132 b^7 + 24 b^5) a^6 + 6 (145 b^{18} + 1092 b^{16} + 1908 b^{14} - 2920 b^{12} \\
& - 3162 b^{10} - 1740 b^8 - 294 b^6) a^5 + 5 (252 b^{17} + 1380 b^{15} + 1884 b^{13} - 762 b^{11} - 96 b^9 \\
& + 24 b^7) a^4 + 4 (145 b^{16} + 600 b^{14} + 93 b^{12} + 198 b^{10} + 3 b^8) a^3 + 3 (74 b^{15} + 162 b^{13} \\
& - 102 b^{11} - 18 b^9) a^2 + 2 (30 b^{14} + 6 b^{12} + 15 b^{10}) a + 8 b^{13} - 6 b^{11}
\end{aligned}$$

#Divide Dar3 by a^{31}

$$collect\left(\frac{Dar3}{a^{31}}, a\right)$$

$$\begin{aligned}
& \frac{22 b^{10}}{a^{10}} + \frac{-126 b^{11} + 168 b^9}{a^{11}} + \frac{300 b^{12} + 120 b^{10} + 600 b^8}{a^{12}} \\
& + \frac{-342 b^{13} - 1938 b^{11} + 3078 b^9 + 1406 b^7}{a^{13}} \\
& + \frac{54 b^{14} + 3564 b^{12} + 1674 b^{10} + 10800 b^8 + 2610 b^6}{a^{14}} \\
& + \frac{408 b^{15} - 1632 b^{13} - 12954 b^{11} + 32028 b^9 + 23460 b^7 + 4284 b^5}{a^{15}} \\
& + \frac{-4704 b^{16} - 27840 b^{14} - 50592 b^{12} - 46720 b^{10} + 30528 b^8 + 17472 b^6 + 2320 b^4}{a^{16}} \\
& + \frac{1}{a^{17}} (360 b^{17} - 1980 b^{15} - 37080 b^{13} - 120150 b^{11} + 50100 b^9 + 70380 b^7 + 20700 b^5 \\
& + 1110 b^3) \\
& + \frac{1}{a^{18}} (42 b^2 (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} - 8307 b^8 - 1172 b^6 + 636 b^4 \\
& + 200 b^2 + 10)) + \frac{1}{a^{19}} (-234 b^{19} - 1248 b^{17} - 32136 b^{15} - 188760 b^{13} - 447408 b^{11} \\
& - 107484 b^9 + 43420 b^7 + 24492 b^5 + 2106 b^3 + 104 b) + \frac{1}{a^{20}} (180 b^{20} + 2376 b^{18} \\
& - 37944 b^{16} - 316224 b^{14} - 808992 b^{12} - 818784 b^{10} - 299052 b^8 - 35040 b^6 + 1116 b^4 \\
& + 72 b^2 + 12) - \frac{1}{a^{21}} (66 b (b^{20} + 17 b^{18} + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} \\
& + 5736 b^8 + 1335 b^6 + 127 b^4 + 17 b^2 + 1)) + \frac{1}{a^{22}} (10 b^2 (b^{20} + 6 b^{18} + 93 b^{16} \\
& - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 - 26352 b^6 - 3162 b^4 + 198 b^2 + 15)) \\
& + \frac{1}{a^{23}} (72 b^{21} + 1458 b^{19} + 16956 b^{17} + 30060 b^{15} - 74412 b^{13} - 309744 b^{11} - 130680 b^9
\end{aligned} \tag{2}$$

$$\begin{aligned}
& - 22248 b^7 - 864 b^5 - 162 b^3) + \frac{1}{a^{24}} (240 b^{20} + 4800 b^{18} + 15264 b^{16} - 28128 b^{14} \\
& - 199368 b^{12} - 210816 b^{10} - 92688 b^8 - 13920 b^6 + 24 b^4) \\
& + \frac{518 b^{19} + 9660 b^{17} + 32844 b^{15} + 23380 b^{13} - 56070 b^{11} - 17304 b^9 - 924 b^7 + 168 b^5}{a^{25}} \\
& + \frac{870 b^{18} + 6552 b^{16} + 11448 b^{14} - 17520 b^{12} - 18972 b^{10} - 10440 b^8 - 1764 b^6}{a^{26}} \\
& + \frac{1260 b^{17} + 6900 b^{15} + 9420 b^{13} - 3810 b^{11} - 480 b^9 + 120 b^7}{a^{27}} \\
& + \frac{580 b^{16} + 2400 b^{14} + 372 b^{12} + 792 b^{10} + 12 b^8}{a^{28}} + \frac{222 b^{15} + 486 b^{13} - 306 b^{11} - 54 b^9}{a^{29}} \\
& + \frac{60 b^{14} + 12 b^{12} + 30 b^{10}}{a^{30}} + \frac{8 b^{13} - 6 b^{11}}{a^{31}}
\end{aligned}$$

#Applies the change of variable

$$sub := \{b = L \cdot a\}$$

$$sub := \{b = L a\} \quad (3)$$

$$\begin{aligned}
& collect\left(subs\left(sub, \frac{Dar3}{a^{31}}\right), a\right) \\
& 10 L^{22} - 66 L^{21} + 180 L^{20} - 234 L^{19} + 42 L^{18} + 360 L^{17} - 4704 L^{16} + 408 L^{15} + 54 L^{14} - 342 L^{13} \quad (4) \\
& + 300 L^{12} - 126 L^{11} + 22 L^{10} + \frac{1}{a^2} (72 L^{21} + 60 L^{20} - 1122 L^{19} + 2376 L^{18} - 1248 L^{17} \\
& - 24360 L^{16} - 1980 L^{15} - 27840 L^{14} - 1632 L^{13} + 3564 L^{12} - 1938 L^{11} + 120 L^{10} + 168 L^9) \\
& + \frac{1}{a^4} (240 L^{20} + 1458 L^{19} + 930 L^{18} - 8382 L^{17} - 37944 L^{16} - 32136 L^{15} - 162204 L^{14} \\
& - 37080 L^{13} - 50592 L^{12} - 12954 L^{11} + 1674 L^{10} + 3078 L^9 + 600 L^8) + \frac{1}{a^6} (518 L^{19} \\
& + 4800 L^{18} + 16956 L^{17} - 29200 L^{16} - 88110 L^{15} - 316224 L^{14} - 188760 L^{13} - 368928 L^{12} \\
& - 120150 L^{11} - 46720 L^{10} + 32028 L^9 + 10800 L^8 + 1406 L^7) + \frac{1}{a^8} (870 L^{18} + 9660 L^{17} \\
& + 15264 L^{16} + 30060 L^{15} - 249210 L^{14} - 378576 L^{13} - 808992 L^{12} - 447408 L^{11} \\
& - 348894 L^{10} + 50100 L^9 + 30528 L^8 + 23460 L^7 + 2610 L^6) + \frac{1}{a^{10}} (1260 L^{17} \\
& + 6552 L^{16} + 32844 L^{15} - 28128 L^{14} - 74412 L^{13} - 682320 L^{12} - 771672 L^{11} - 818784 L^{10} \\
& - 107484 L^9 - 49224 L^8 + 70380 L^7 + 17472 L^6 + 4284 L^5) + \frac{1}{a^{12}} (580 L^{16} + 6900 L^{15} \\
& + 11448 L^{14} + 23380 L^{13} - 199368 L^{12} - 309744 L^{11} - 674160 L^{10} - 378576 L^9 - 299052 L^8
\end{aligned}$$

$$\begin{aligned}
& + 43420 L^7 + 26712 L^6 + 20700 L^5 + 2320 L^4) + \frac{1}{a^{14}} (222 L^{15} + 2400 L^{14} + 9420 L^{13} \\
& - 17520 L^{12} - 56070 L^{11} - 210816 L^{10} - 130680 L^9 - 263520 L^8 - 88110 L^7 - 35040 L^6 \\
& + 24492 L^5 + 8400 L^4 + 1110 L^3) + \frac{1}{a^{16}} (60 L^{14} + 486 L^{13} + 372 L^{12} - 3810 L^{11} \\
& - 18972 L^{10} - 17304 L^9 - 92688 L^8 - 22248 L^7 - 31620 L^6 - 8382 L^5 + 1116 L^4 + 2106 L^3 \\
& + 420 L^2) + \frac{1}{a^{18}} (8 L^{13} + 12 L^{12} - 306 L^{11} + 792 L^{10} - 480 L^9 - 10440 L^8 - 924 L^7 \\
& - 13920 L^6 - 864 L^5 + 1980 L^4 - 1122 L^3 + 72 L^2 + 104 L) + \frac{1}{a^{20}} (-6 L^{11} + 30 L^{10} \\
& - 54 L^9 + 12 L^8 + 120 L^7 - 1764 L^6 + 168 L^5 + 24 L^4 - 162 L^3 + 150 L^2 - 66 L + 12)
\end{aligned}$$

#So $\frac{Dar3}{a^{31}}$ tends to

$$\begin{aligned}
& 10 L^{22} - 66 L^{21} + 180 L^{20} - 234 L^{19} + 42 L^{18} + 360 L^{17} - 4704 L^{16} + 408 L^{15} + 54 L^{14} - 342 L^{13} \\
& + 300 L^{12} - 126 L^{11} + 22 L^{10} :
\end{aligned}$$

#Similarly

$Dbr3 := \text{diff}(r3, b)$

$$\begin{aligned}
Dbr3 := & 12 b^{11} + 2 b (b^{20} + 6 b^{18} + 93 b^{16} - 2920 b^{14} - 24921 b^{12} - 68232 b^{10} - 67416 b^8 \\
& - 26352 b^6 - 3162 b^4 + 198 b^2 + 15) a^{10} + 6 b (b^{16} - 580 b^{14} - 3862 b^{12} - 8784 b^{10} \\
& - 8307 b^8 - 1172 b^6 + 636 b^4 + 200 b^2 + 10) a^{14} + (420 b^{13} + 72 b^{11} + 150 b^9) a^2 \\
& + (104 b^{12} - 66 b^{10}) a + 10 a^{22} b^9 + (-66 b^{10} + 72 b^8) a^{21} + (180 b^{11} + 60 b^9 \\
& + 240 b^7) a^{20} + (-234 b^{12} - 1122 b^{10} + 1458 b^8 + 518 b^6) a^{19} + (42 b^{13} + 2376 b^{11} \\
& + 930 b^9 + 4800 b^7 + 870 b^5) a^{18} + (360 b^{14} - 1248 b^{12} - 8382 b^{10} + 16956 b^8 + 9660 b^6 \\
& + 1260 b^4) a^{17} + (-4704 b^{15} - 24360 b^{13} - 37944 b^{11} - 29200 b^9 + 15264 b^7 + 6552 b^5 \\
& + 580 b^3) a^{16} + (408 b^{16} - 1980 b^{14} - 32136 b^{12} - 88110 b^{10} + 30060 b^8 + 32844 b^6 \\
& + 6900 b^4 + 222 b^2) a^{15} + (-342 b^{18} - 1632 b^{16} - 37080 b^{14} - 188760 b^{12} - 378576 b^{10} \\
& - 74412 b^8 + 23380 b^6 + 9420 b^4 + 486 b^2 + 8) a^{13} + (300 b^{19} + 3564 b^{17} - 50592 b^{15} \\
& - 368928 b^{13} - 808992 b^{11} - 682320 b^9 - 199368 b^7 - 17520 b^5 + 372 b^3 + 12 b) a^{12} \\
& + (168 b^{20} + 3078 b^{18} + 32028 b^{16} + 50100 b^{14} - 107484 b^{12} - 378576 b^{10} - 130680 b^8 \\
& - 17304 b^6 - 480 b^4 - 54 b^2) a^9 + (600 b^{19} + 10800 b^{17} + 30528 b^{15} - 49224 b^{13} \\
& - 299052 b^{11} - 263520 b^9 - 92688 b^7 - 10440 b^5 + 12 b^3) a^8 + (1406 b^{18} + 23460 b^{16} \\
& + 70380 b^{14} + 43420 b^{12} - 88110 b^{10} - 22248 b^8 - 924 b^6 + 120 b^4) a^7 + (2610 b^{17} \\
& + 17472 b^{15} + 26712 b^{13} - 35040 b^{11} - 31620 b^9 - 13920 b^7 - 1764 b^5) a^6 + (4284 b^{16} \\
& + 20700 b^{14} + 24492 b^{12} - 8382 b^{10} - 864 b^8 + 168 b^6) a^5 + (2320 b^{15} + 8400 b^{13}
\end{aligned} \tag{5}$$

$$\begin{aligned}
& + 1116 b^{11} + 1980 b^9 + 24 b^7) a^4 + (1110 b^{14} + 2106 b^{12} - 1122 b^{10} - 162 b^8) a^3 \\
& + b^2 (20 b^{19} + 108 b^{17} + 1488 b^{15} - 40880 b^{13} - 299052 b^{11} - 682320 b^9 - 539328 b^7 \\
& - 158112 b^5 - 12648 b^3 + 396 b) a^{10} - 6 b (20 b^{19} + 306 b^{17} + 2032 b^{15} + 18690 b^{13} \\
& + 68832 b^{11} + 116920 b^9 + 45888 b^7 + 8010 b^5 + 508 b^3 + 34 b) a^{11} - 6 (b^{20} + 17 b^{18} \\
& + 127 b^{16} + 1335 b^{14} + 5736 b^{12} + 11692 b^{10} + 5736 b^8 + 1335 b^6 + 127 b^4 + 17 b^2 + 1) a^{11} \\
& + 3 b^2 (16 b^{15} - 8120 b^{13} - 46344 b^{11} - 87840 b^9 - 66456 b^7 - 7032 b^5 + 2544 b^3 \\
& + 400 b) a^{14}
\end{aligned}$$

$$collect\left(subs\left(sub, \frac{Dbr3}{a^{31}}\right), a\right)$$

$$\begin{aligned}
& 22 L^{21} - 126 L^{20} + 300 L^{19} - 342 L^{18} + 54 L^{17} + 408 L^{16} - 4704 L^{15} + 360 L^{14} + 42 L^{13} - 234 L^{12} \quad (6) \\
& + 180 L^{11} - 66 L^{10} + 10 L^9 + \frac{1}{a^2} (168 L^{20} + 120 L^{19} - 1938 L^{18} + 3564 L^{17} - 1632 L^{16} \\
& - 27840 L^{15} - 1980 L^{14} - 24360 L^{13} - 1248 L^{12} + 2376 L^{11} - 1122 L^{10} + 60 L^9 + 72 L^8) \\
& + \frac{1}{a^4} (600 L^{19} + 3078 L^{18} + 1674 L^{17} - 12954 L^{16} - 50592 L^{15} - 37080 L^{14} - 162204 L^{13} \\
& - 32136 L^{12} - 37944 L^{11} - 8382 L^{10} + 930 L^9 + 1458 L^8 + 240 L^7) + \frac{1}{a^6} (1406 L^{18} \\
& + 10800 L^{17} + 32028 L^{16} - 46720 L^{15} - 120150 L^{14} - 368928 L^{13} - 188760 L^{12} - 316224 L^{11} \\
& - 88110 L^{10} - 29200 L^9 + 16956 L^8 + 4800 L^7 + 518 L^6) + \frac{1}{a^8} (2610 L^{17} + 23460 L^{16} \\
& + 30528 L^{15} + 50100 L^{14} - 348894 L^{13} - 447408 L^{12} - 808992 L^{11} - 378576 L^{10} - 249210 L^9 \\
& + 30060 L^8 + 15264 L^7 + 9660 L^6 + 870 L^5) + \frac{1}{a^{10}} (4284 L^{16} + 17472 L^{15} + 70380 L^{14} \\
& - 49224 L^{13} - 107484 L^{12} - 818784 L^{11} - 771672 L^{10} - 682320 L^9 - 74412 L^8 - 28128 L^7 \\
& + 32844 L^6 + 6552 L^5 + 1260 L^4) + \frac{1}{a^{12}} (2320 L^{15} + 20700 L^{14} + 26712 L^{13} + 43420 L^{12} \\
& - 299052 L^{11} - 378576 L^{10} - 674160 L^9 - 309744 L^8 - 199368 L^7 + 23380 L^6 + 11448 L^5 \\
& + 6900 L^4 + 580 L^3) + \frac{1}{a^{14}} (1110 L^{14} + 8400 L^{13} + 24492 L^{12} - 35040 L^{11} - 88110 L^{10} \\
& - 263520 L^9 - 130680 L^8 - 210816 L^7 - 56070 L^6 - 17520 L^5 + 9420 L^4 + 2400 L^3 \\
& + 222 L^2) + \frac{1}{a^{16}} (420 L^{13} + 2106 L^{12} + 1116 L^{11} - 8382 L^{10} - 31620 L^9 - 22248 L^8 \\
& - 92688 L^7 - 17304 L^6 - 18972 L^5 - 3810 L^4 + 372 L^3 + 486 L^2 + 60 L) \\
& + \frac{1}{a^{18}} (104 L^{12} + 72 L^{11} - 1122 L^{10} + 1980 L^9 - 864 L^8 - 13920 L^7 - 924 L^6 - 10440 L^5
\end{aligned}$$

$$- 480 L^4 + 792 L^3 - 306 L^2 + 12 L + 8) + \frac{1}{a^{20}} (12 L^{11} - 66 L^{10} + 150 L^9 - 162 L^8 \\ + 24 L^7 + 168 L^6 - 1764 L^5 + 120 L^4 + 12 L^3 - 54 L^2 + 30 L - 6)$$

#So $\frac{Dbr3}{a^{31}}$ tends **to**

$$22 L^{21} - 126 L^{20} + 300 L^{19} - 342 L^{18} + 54 L^{17} + 408 L^{16} - 4704 L^{15} + 360 L^{14} + 42 L^{13} - 234 L^{12} \\ + 180 L^{11} - 66 L^{10} + 10 L^9 :$$

#Then L satisfies

$$p := L \cdot (22 L^{21} - 126 L^{20} + 300 L^{19} - 342 L^{18} + 54 L^{17} + 408 L^{16} - 4704 L^{15} + 360 L^{14} + 42 L^{13} \\ - 234 L^{12} + 180 L^{11} - 66 L^{10} + 10 L^9) - 10 L^{22} - 66 L^{21} + 180 L^{20} - 234 L^{19} + 42 L^{18} + 360 L^{17} \\ - 4704 L^{16} + 408 L^{15} + 54 L^{14} - 342 L^{13} + 300 L^{12} - 126 L^{11} + 22 L^{10} \\ p := L (22 L^{21} - 126 L^{20} + 300 L^{19} - 342 L^{18} + 54 L^{17} + 408 L^{16} - 4704 L^{15} + 360 L^{14} + 42 L^{13} \quad (7) \\ - 234 L^{12} + 180 L^{11} - 66 L^{10} + 10 L^9) - 10 L^{22} - 66 L^{21} + 180 L^{20} - 234 L^{19} + 42 L^{18} \\ + 360 L^{17} - 4704 L^{16} + 408 L^{15} + 54 L^{14} - 342 L^{13} + 300 L^{12} - 126 L^{11} + 22 L^{10}$$

simplify(p)

$$12 L^{10} \left(L^{12} - 16 L^{11} + 40 L^{10} - 48 L^9 + 8 L^8 + 64 L^7 - 784 L^6 + 64 L^5 + 8 L^4 - 48 L^3 + 40 L^2 \quad (8) \right. \\ \left. - 16 L + \frac{8}{3} \right)$$

eval(p, L=1)

$$-8212 \quad (9)$$

#Therefore L is diferent for 1

#For check that L is not 0, suppose the opposite.

$$\begin{aligned}
 & \text{expand}\left(\frac{r^3}{b^{10} \cdot a^{22}}\right) \\
 & 1 + \frac{b^2}{a^{22}} - \frac{6b}{a} + \frac{1092}{a^6 b^4} + \frac{4692}{a^7 b^3} - \frac{3516}{a^8 b^2} - \frac{8268}{a^9 b} - \frac{70152b}{a^{11}} - \frac{68232b^2}{a^{12}} - \frac{8268b^3}{a^{13}} \\
 & - \frac{3516b^4}{a^{14}} + \frac{4692b^5}{a^{15}} + \frac{1092b^6}{a^{16}} + \frac{252b^7}{a^{17}} + \frac{145}{a^6 b^6} + \frac{1380}{a^7 b^5} + \frac{1908}{a^8 b^4} + \frac{3340}{a^9 b^3} \\
 & - \frac{24921}{a^{10} b^2} - \frac{34416}{a^{11} b} - \frac{34416b}{a^{13}} - \frac{24921b^2}{a^{14}} + \frac{3340b^3}{a^{15}} + \frac{1908b^4}{a^{16}} + \frac{1380b^5}{a^{17}} \\
 & + \frac{145b^6}{a^{18}} + \frac{74}{a^7 b^7} + \frac{600}{a^8 b^6} + \frac{1884}{a^9 b^5} - \frac{2920}{a^{10} b^4} - \frac{8010}{a^{11} b^3} - \frac{26352}{a^{12} b^2} - \frac{14520}{a^{13} b} \\
 & - \frac{8010b}{a^{15}} - \frac{2920b^2}{a^{16}} + \frac{1884b^3}{a^{17}} + \frac{600b^4}{a^{18}} + \frac{74b^5}{a^{19}} + \frac{30}{a^8 b^8} + \frac{162}{a^9 b^7} + \frac{93}{a^{10} b^6} \\
 & - \frac{762}{a^{11} b^5} - \frac{3162}{a^{12} b^4} - \frac{2472}{a^{13} b^3} - \frac{11586}{a^{14} b^2} - \frac{2472}{a^{15} b} - \frac{762b}{a^{17}} + \frac{93b^2}{a^{18}} + \frac{162b^3}{a^{19}} + \frac{30b^4}{a^{20}} \\
 & + \frac{8}{a^9 b^9} + \frac{6}{a^{10} b^8} - \frac{102}{a^{11} b^7} + \frac{198}{a^{12} b^6} - \frac{96}{a^{13} b^5} - \frac{1740}{a^{14} b^4} - \frac{132}{a^{15} b^3} - \frac{1740}{a^{16} b^2} - \frac{96}{a^{17} b} \\
 & - \frac{102b}{a^{19}} + \frac{6b^2}{a^{20}} + \frac{8b^3}{a^{21}} + \frac{1}{a^{10} b^{10}} - \frac{6}{a^{11} b^9} + \frac{15}{a^{12} b^8} - \frac{18}{a^{13} b^7} + \frac{3}{a^{14} b^6} + \frac{24}{a^{15} b^5} \\
 & - \frac{294}{a^{16} b^4} + \frac{24}{a^{17} b^3} + \frac{3}{a^{18} b^2} - \frac{18}{a^{19} b} - \frac{6b}{a^{21}} + \frac{15b^2}{a^2} - \frac{18b^3}{a^3} + \frac{3b^4}{a^4} + \frac{24b^5}{a^5} \\
 & - \frac{294b^6}{a^6} + \frac{24b^7}{a^7} + \frac{3b^8}{a^8} - \frac{18b^9}{a^9} - \frac{6b^{11}}{a^{11}} + \frac{b^{12}}{a^{12}} + \frac{8}{ab} - \frac{102b}{a^3} + \frac{198b^2}{a^4} \\
 & - \frac{96b^3}{a^5} - \frac{1740b^4}{a^6} - \frac{132b^5}{a^7} - \frac{1740b^6}{a^8} - \frac{96b^7}{a^9} + \frac{198b^8}{a^{10}} - \frac{102b^9}{a^{11}} + \frac{6b^{10}}{a^{12}} \\
 & + \frac{8b^{11}}{a^{13}} + \frac{30}{a^2 b^2} + \frac{162}{a^3 b} - \frac{762b}{a^5} - \frac{3162b^2}{a^6} - \frac{2472b^3}{a^7} - \frac{11586b^4}{a^8} - \frac{2472b^5}{a^9} \\
 & - \frac{3162b^6}{a^{10}} - \frac{762b^7}{a^{11}} + \frac{93b^8}{a^{12}} + \frac{162b^9}{a^{13}} + \frac{30b^{10}}{a^{14}} + \frac{74}{a^3 b^3} + \frac{600}{a^4 b^2} + \frac{1884}{a^5 b} \\
 & - \frac{8010b}{a^7} - \frac{26352b^2}{a^8} - \frac{14520b^3}{a^9} - \frac{26352b^4}{a^{10}} - \frac{8010b^5}{a^{11}} - \frac{2920b^6}{a^{12}} + \frac{1884b^7}{a^{13}} \\
 & + \frac{600b^8}{a^{14}} + \frac{74b^9}{a^{15}} + \frac{145}{a^4 b^4} + \frac{1380}{a^5 b^3} + \frac{1908}{a^6 b^2} + \frac{3340}{a^7 b} - \frac{34416b}{a^9} - \frac{67416b^2}{a^{10}} \\
 & - \frac{34416b^3}{a^{11}} - \frac{24921b^4}{a^{12}} + \frac{3340b^5}{a^{13}} + \frac{1908b^6}{a^{14}} + \frac{1380b^7}{a^{15}} + \frac{145b^8}{a^{16}} + \frac{252}{a^5 b^5}
 \end{aligned} \tag{10}$$

$$\begin{aligned}
& + \frac{15 b^{10}}{a^{10}} + \frac{6}{a^2} + \frac{93}{a^4} - \frac{2920}{a^6} - \frac{24921}{a^8} - \frac{68232}{a^{10}} - \frac{67416}{a^{12}} - \frac{26352}{a^{14}} - \frac{3162}{a^{16}} \\
& + \frac{198}{a^{18}} + \frac{15}{a^{20}}
\end{aligned}$$

#Aplying the limit, we obtains $l=0$, a contratiction.